

LESSON 4 : PERCENTAGES II (A)

Checkpoint 1

The amount of rainfall of a city was 2500 mm in 2019. The amount of rainfall in 2020 was increased by 12%, and in 2021, it was decreased by 15%. Find the amount of rainfall in 2021.

∴ The amount of rainfall in 2021,
 $2500 (1 + 12\%) (1 - 15\%)$
 $= 2094.4 \text{ mm}$

Rainfall (mm)	
2019	2500
2020	2500 (1 + 12%)
2021	2500 (1 + 12%) (1 - 15%)

1. Find the new value in each of the following.

- (a) 200 increases by 30% and then increases by 40%.
- (b) 8 m decreases by 50% and then decreases by 10%.
- (c) \$1 600 decreases by 25% and then increases by 60%.

◆ 2. The amount of rainfall of a city was 2 200 mm in 2010. The amount of rainfall in 2011 was decreased by 18%, and in 2012, it was increased by 2%. Find the amount of rainfall in 2012.

3. There are 200 books on a bookshelf, of which 60% are fiction and the rest are non-fiction. If the numbers of fiction and non-fiction books are increased by 10% and 20% respectively, find
- (a) the number of fiction books now,
 - (b) the number of non-fiction books now,
 - (c) the increase in the number of books on the bookshelf.

Checkpoint 2

Tony's monthly salary was increased by 4% and 3.5% in last two years respectively. If Tony's monthly salary is \$13 455 in 2020, find his monthly salary 2 years ago (i.e. in 2018).

Salary (\$)	
2018	x
2019	$x(1 + 4\%)$
2020	$x(1 + 4\%)(1 + 3.5\%) = \$13\,455$

- ◆ 4. A table is worth \$697 now. Its current value is 18% lower than that of last year. Last year, its value was 15% lower than that of 2 years ago. Find the value of the table 2 years ago.
5. When a number is first increased by 15% and then decreased by 30%, the result is 241.5. What is the original value of the number?
6. If a number is increased by 25% and then decreased by 10%, it becomes 99. Find the original number.
7. The monthly salary of Mr. Lee was increased by 12% and 25% in the last two years. If his current monthly salary is \$17 500, what was his monthly salary two years ago?
8. Find the original value in each of the following.
- (a) After increasing by 6% and then increasing by 7%, the new value is 90 736.
 - (b) After increasing by 30% and then decreasing by 20%, the new value is 208 g.
 - (c) After decreasing by 40% and then increasing by 60%, the new value is 76.8 cm^3 .

- ◆ 9. The price of a car is \$125 000. The price is first decreased by 15% and then further decreased by 8%. Find the overall percentage change in the price of the car.
10. The volume of water in the container is 1.25L. The volume is first increased by 22% and then further decreased by 30%. Find the overall percentage change in the volume of water.
11. The tax revenue of a government increased by 20%, 10%, 35% respectively in the first three quarters of 2021, and then decreased by 45% in the fourth quarter of the year. What was the overall percentage change in the quarterly tax revenue over the whole year 2021?
12. If a number is decreased by 90% and then increased by $x\%$, the overall percentage change is -83% . Find x .

13. The length, the width and the height of a rectangular block are 15 cm, 8 cm, and 18 cm respectively. If the length increases by 20% and the width increases by 15%, but the volume decreases by 10%, find the percentage change in height.
14. The length, the width and the height of a rectangular block are 10 cm, 15 cm, and 21 cm respectively. If the length decreases by 15% and the width increases by 16%, but the volume increases by 6%, find the percentage change in height.

%

15. The radius r cm of a circle is decreased by 20%.
- (a) Express the original area of the circle in terms of r .
 - (b) Express the new area of the circle in terms of r .
 - (c) Find the percentage change in the area of the circle.

Hints:

$$\begin{array}{rcl} \frac{3x - x}{x} & & \frac{3x^2 - x^2}{x^2} \\ = \frac{2x}{x} & & = \frac{2x^2}{x^2} \\ = 2 & & = 2 \end{array}$$

16. The length of a square is l cm. If the length is first increased by 20% and then further increased by 20%,
- (a) express the new area of the square in terms of l ,
 - (b) find the percentage change in the area of the square.

17. Evelyn's pocket money was increased by 10% last month, but is decreased by 10% this month. What is the overall percentage change of her pocket money?
[Hint: Let $\$x$ be Evelyn's original pocket money.]
18. The length and the width of a rectangle are l cm and w cm respectively. If the length is increased by 10% and the width is decreased by 30%,
- (a) express the new area of the rectangle in terms of l and w ,
 - (b) find the percentage change in the area of the rectangle.
19. The angle of a sector is decreased by 20% but its radius is increased by $k\%$. If the arc length of the sector remains unchanged, find the value of k .

- ◆ 20. In a normal day, Helen spends 37.5% of the time sleeping and 10% of the remaining time studying.
- (a) What is the amount of time used for other purposes in a normal day?
 - (b) An examination is approaching, so Helen has to spend more time on studying. She has decided to reduce the sleeping time and other purposes by 5% respectively.
 - (i) Find the percentage change in Helen's studying time.
 - (ii) Furthermore, she increases the reading speed by 4%. Find the percentage increase in the amount of reading materials she can read.

21. The operating cost of a company is composed of three parts:

Advertising	\$10 000	Rent	\$40 000	Wage	\$200 000
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If the advertising cost increases by 10%, the rent decreases by 20% and the wage increases by 5%, find the percentage change in the operating cost of the company.

22. Last week, the number of books sold in bookstores P , Q and R were 150, 240 and 320 respectively. As compared to last week, the number of books sold this week in bookstore P decreases by 2%, that in bookstore Q increases by 5% and that in bookstore R remains unchanged. Find the percentage change in the total number of books sold in the three bookstores, correct to 3 significant figures.
23. Last week, the numbers of male customers and the female customers in a shopping mall were in the ratio 1 : 3. This week, the number of male customers increases by 16% while the number of female customers decreases by 12%. Is there an increase in the total number of customers as compared to last week? Explain your answer.
24. The numbers of tourists of three cities A , B and C in 2020 are listed below:

City	A	B	C
Number of tourists	2 500	900	1 600

In 2021, the numbers of tourists of cities B and C increased by 40% and 20% respectively. If the total number of tourists of the three cities increased by 4%, find the percentage change in the number of tourists of city A .

25. There are some apples, oranges and peaches in a box. The numbers of apples, oranges and peaches are in the ratio 2 : 5 : 3. If the number of apples decreases by 40%, the number of oranges increases by 15% and the number of peaches decreases by 5%, find the percentage change in the number of fruits in the box.
26. If the height of a right cylinder increases by 60% and its curved surface area remains unchanged, find the percentage change in the volume of the cylinder.